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Physics in Exile

Max Dresden entered the crowded seminar room, all eyes on him as he took his place alone in front of the chalkboard. Dresden was a physicist at the University of Kansas, and, during his visit to Princeton's Institute for Advanced Study in 1952, he had volunteered to give a talk on the fascinating new work by David Bohm. Dresden was eager to hear what his audience thought of Bohm's work: the "Princetitute" was home to some of the finest minds in all of physics, including Einstein himself—though, as Dresden looked out at the room, Einstein's unruly white hair was nowhere to be seen.

Dresden's students had brought Bohm's paper to his attention, and he had initially dismissed their questions by citing von Neumann's famous proof that the Copenhagen interpretation was the only way of understanding quantum physics. But, after repeated pestering, Dresden finally looked at Bohm's paper, and he was surprised by what he found. Bohm had discovered a totally new way to interpret quantum physics. Rather than refusing to answer questions about the quantum world, as the Copenhagen interpretation did, Bohm's interpretation depicted a world of subatomic particles that existed whether or not anyone was looking at them, particles with definite positions at all times. These particles, in turn, each had "pilot waves" that determined their motion, which also behaved in an orderly and predictable fashion. Somehow, Bohm had found a way to tame the chaotic and unknowable world of the quantum—and he had done it without sacrificing accuracy, because Bohm's theory was mathematically equivalent